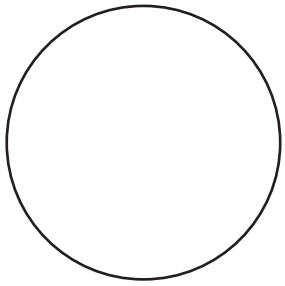
0580_w23_qp_31_Q2

2 **(a)** The diagram shows a circle.



NOT TO SCALE

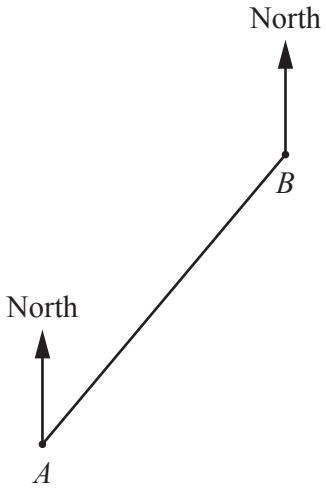
(i) The diameter of this circle is 168 mm.

Write down the radius of this circle.

..... mm [1]

(ii) On the diagram, draw a chord of this circle. [1]

(b) The scale drawing shows the position of ship *A* and the position of ship *B*.
The scale is 1 cm represents 6 km.



Scale : 1 cm to 6 km

Another ship, *C*, is 45 km from ship *B* on a bearing of 124°.

(i) On the scale drawing, mark the position of ship *C*. [2]

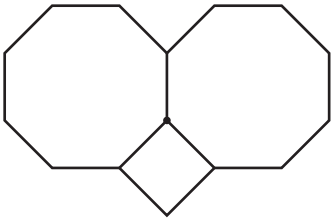
(ii) Find the actual distance of ship *C* from ship *A*.

..... km [2]

(c) **(i)** Show that the interior angle of a regular octagon is 135°.

[1]

(ii)

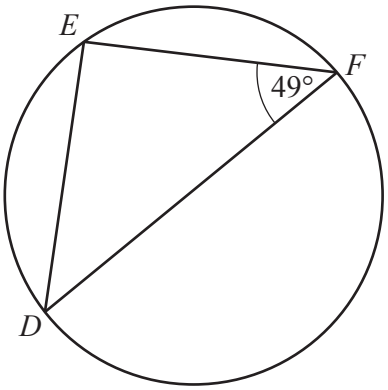


NOT TO SCALE

Show that two regular octagons and a square meet at a point without any gaps.

[1]

(d)



NOT TO SCALE

The diagram shows points *D*, *E* and *F* on the circumference of a circle.
DF is a diameter of the circle.

Find angle *EDF*.

Angle *EDF* = ... [2]